Stress Zone Change Detection — Technical Documentation

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Overview

The project turns repeat drone or aircraft flights over the same fields into maps of where crop stress is spreading, persisting, or recovering across a season. Each flight yields a vegetation index raster (NDVI, NDRE, etc.); consecutive rasters are co-registered, differenced, and classified into change classes per pixel and per stress patch.

Scope

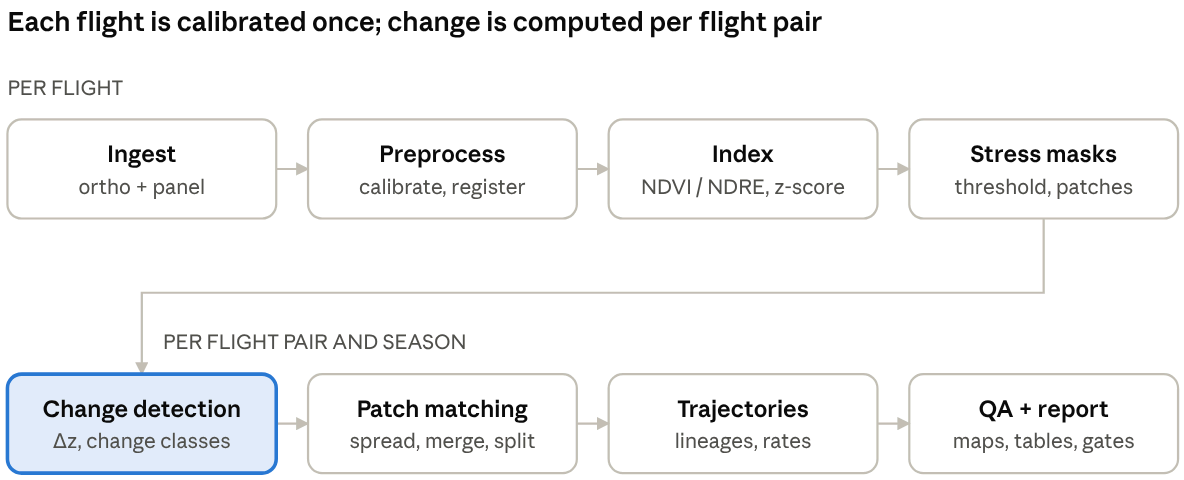
- In: multispectral orthomosaics from repeat flights, field boundaries, optional ground-truth scouting points.
- Out: per-date stress masks, pairwise change rasters, patch-level change vectors, season trajectory tables, and summary maps/reports.
- Not in scope: flight planning, orthomosaic stitching, and agronomic diagnosis of stress cause (water, nitrogen, disease).

Key outputs

Output	Format	Granularity
Vegetation index rasters	Cloud-Optimized GeoTIFF (float32)	Per flight
Stress masks	COG (uint8)	Per flight
Change rasters (ΔVI, change class)	COG	Per flight pair
Stress patches + change attributes	GeoPackage	Per flight pair
Season trajectories	Parquet / CSV	Per patch, per zone
QA report	HTML / PDF	Per run

Architecture

The pipeline has two halves: per-flight stages that run once when a flight lands, and pair/season stages that rerun whenever a new flight joins the series.



processing pipeline · 8 stages

Each stage reads from and writes to `derived/` with a JSON sidecar; a stage reruns only when its inputs or config hash change. Implementation stack: Python 3.11, rasterio/GDAL, numpy, scikit-image (morphology, labelling, registration), geopandas/shapely (patches), xarray for the season stack, and a Typer-based `szcd` CLI. Stages parallelize across fields and dates.

Data inputs

Change detection is only as good as flight consistency: same sensor, similar sun angle, and a calibration panel every flight.

Flight requirements

Parameter	Requirement	Why
Sensor	Same multispectral camera all season (e.g. Red, Red Edge, NIR bands)	Band response differences masquerade as change
Ground sample distance	Within ±20% across flights; target 3–10 cm	Resampling to a common grid must not blur patches

Parameter	Requirement	Why
Time of day	Within ±2 h of solar noon	Limits shadow and BRDF effects
Radiometric reference	Reflectance panel captured pre- and post-flight; downwelling light sensor if available	Converts DN to reflectance
Revisit interval	7–14 days during active growth	Captures spread before it saturates
Ground control	⩾ 5 GCPs or RTK/PPK positioning	Sub-pixel co-registration

Ancillary inputs

- Field boundaries (GeoJSON/GPKG, one feature per field, with `field_id`).
- Optional masks: headlands, tramlines, irrigation infrastructure, known non-crop areas.
- Optional scouting points with date, location, and observed stress label for validation.

Directory layout

```
data/
  raw/<field_id>/<YYYY-MM-DD>/ortho_multispectral.tif
  raw/<field_id>/<YYYY-MM-DD>/panel.json
  ancillary/fields.gpkg
  ancillary/masks/<field_id>.gpkg
  ancillary/scouting.csv
derived/
  vi/<field_id>/<date>_<index>.tif
  stress/<field_id>/<date>_mask.tif
  change/<field_id>/<date_a>__<date_b>_{delta,class}.tif
  patches/<field_id>/<date_a>__<date_b>.gpkg
  trajectories/<field_id>.parquet
reports/<run_id>/
config/season.yaml
```

Preprocessing

Every flight is brought onto one season reference grid in calibrated reflectance before any index is computed.

1. **Radiometric calibration** — convert DN to surface reflectance using the panel's known reflectance (empirical line) and, where present, the downwelling light sensor for irradiance correction. Reject a flight if panel-derived gain differs from the season median by > 15% without a DLS explanation.
2. **Relative normalization** — optional pseudo-invariant feature (PIF) normalization against the reference flight using stable targets (roads, bare soil margins, panels). Fits a per-band linear model; residual RMSE is logged.
3. **Co-registration** — align each orthomosaic to the reference flight (default: first good flight of the season) using feature matching (ORB/SIFT) or phase correlation, then an affine or thin-plate-spline warp. Accept when RMSE ⩽ 0.5 pixel on check points.
4. **Resampling** — warp to the reference grid (CRS, origin, GSD) with bilinear resampling for reflectance; nearest for masks.
5. **Masking** — apply, in order: outside-field mask, ancillary non-crop mask, shadow mask (low NIR + low visible brightness), saturation mask, soil mask (VI below soil threshold early season). Masked pixels carry nodata (−9999) through all later steps.
6. **Smoothing (optional)** — a small median or Gaussian filter (σ = 1–2 px) to suppress row-level texture before differencing.

Each step writes a JSON sidecar with parameters and quality metrics, used by the QA report.

Vegetation indices and differencing

NDRE is the default for mid–late season because NDVI saturates over dense canopy; NDVI is used early season.

Index	Formula	Best use
NDVI	(NIR − Red) / (NIR + Red)	Early season, sparse canopy
NDRE	(NIR − RedEdge) / (NIR + RedEdge)	Dense canopy, chlorophyll/N stress
GNDVI	(NIR − Green) / (NIR + Green)	Chlorophyll, less saturation than NDVI
SAVI	1.5 × (NIR − Red) / (NIR + Red + 0.5)	High soil exposure

Differencing methods (configurable per run):

- **Simple difference** — $\Delta VI = VI(t_2) - VI(t_1)$. Negative = decline.

- **Relative difference** — $\Delta VI / VI(t_1)$; emphasises change in low-vigour areas.
- **Field-normalized difference** — z-score each date against the field's healthy-canopy distribution first, then difference. Removes whole-field phenology shifts (green-up, senescence) so only *relative* stress change remains. This is the default.

$$z_t(x) = \frac{VI_t(x) - \mathrm{median}_t}{1.4826 \cdot \mathrm{MAD}_t}, \qquad \Delta z(x) = z_{t_2}(x) - z_{t_1}(x)$$

Median and MAD are computed over valid crop pixels in the field. A pixel counts as significant change when $|\Delta z|$ exceeds `change.threshold_z` (default 1.0) and, for simple differences, $|\Delta VI|$ exceeds the sensor noise floor estimated from PIF residuals.

Stress zones and change classification

A pixel is stressed on a date when its field z-score falls below `stress.threshold_z` (default −1.5); change classes come from comparing the two dates' stress states plus the Δz sign.

Stress delineation per date

1. Threshold z-scores into a binary stress mask (alternatives: Otsu on the field histogram, or k-means into 3 vigour zones).
2. Morphological opening then closing (radius `stress.morph_radius_px`, default 2) to drop speckle and close row gaps.
3. Connected-component labelling (8-connectivity); drop patches below `stress.min_area_m2` (default 25 m²).
4. Vectorize to polygons with `patch_id`, area, mean/min z, centroid.

Pixel change classes ($t_1 \rightarrow t_2$)

Class	Code	Rule
Stable healthy	0	not stressed → not stressed
New stress	1	not stressed → stressed, outside any t_1 patch buffer
Spread	2	not stressed → stressed, within `change.spread_buffer_m` (default 5 m) of a t_1 patch

Class	Code	Rule
Persistent	3	stressed → stressed,
Worsening	4	stressed → stressed, $\Delta z \leqslant -$threshold
Recovering	5	stressed → stressed, $\Delta z \geqslant +$threshold
Recovered	6	stressed → not stressed
No data	255	masked on either date

Patch matching across dates

Patches at t_1 and t_2 are linked by spatial overlap (IoU ⩾ 0.1, or centroid within buffer). Each link is labelled:

- **Expanding / shrinking / stable** — area ratio $A_2/A_1 > 1.2$, < 0.8, otherwise stable.
- **Merged** — two or more t_1 patches overlap one t_2 patch.
- **Split** — one t_1 patch overlaps several t_2 patches.
- **Appeared / disappeared** — no match on the other date.

Patch attributes written to the GeoPackage: `area_t1_m2`, `area_t2_m2`, `area_change_pct`, `mean_dz`, `edge_advance_m` (mean outward boundary shift, from distance transforms), `spread_direction_deg`, `event`.

Seasonal trajectory analysis

Pairwise links are chained into patch lineages so each stress zone has one history from first appearance to recovery or harvest.

Lineage building — patch matches form a directed graph across dates; each connected chain gets a `lineage_id`. Merges keep the larger parent's ID; splits create child IDs that reference the parent.

Per-lineage metrics

Metric	Unit	Definition
`onset_date`	date	First flight the lineage is detected
`peak_area_m2` / `peak_date`	m², date	Maximum extent and when

Metric	Unit	Definition
`spread_rate_m2_per_day`	m²/day	Area slope between onset and peak (linear fit)
`edge_advance_m_per_day`	m/day	Mean boundary advance rate
`severity_auc`	z·day	Area under the −z curve over the lineage's life
`recovery_date`	date	First flight with area < 20% of peak, if any
`status_final`	enum	active, recovering, recovered, persistent

Field-level summaries per date — % of crop area in each change class, stressed area total, number of active lineages. These run as time series in the report.

Temporal consistency filter — a single-date stress blip not seen on the dates before or after (and under `trajectory.min_persistence` flights, default 2) is flagged as `transient` and excluded from spread metrics; often a cloud shadow or irrigation timing artefact.

Configuration, CLI and outputs

One YAML file per season drives every stage; each stage can be re-run alone and skips work whose inputs and config hash are unchanged.

`config/season.yaml`

```
season: 2026_corn_north
reference_date: 2026-05-28        # co-registration + normalization target
fields: data/ancillary/fields.gpkg
bands: {green: 2, red: 3, red_edge: 4, nir: 5}
grid: {crs: EPSG:32610, gsd_m: 0.05}
calibration:
  method: empirical_line
  use_dls: true
  max_gain_deviation: 0.15
coregistration:
  method: feature            # feature | phase
  transform: affine          # affine | tps
  max_rmse_px: 0.5
index:
  primary: NDRE
  early_season: {index: NDVI, until: 2026-06-20}
```

```
smoothing: {method: median, size_px: 3}
stress:
  method: zscore             # zscore | otsu | kmeans
  threshold_z: -1.5
  morph_radius_px: 2
  min_area_m2: 25
change:
  method: field_normalized   # simple | relative | field_normalized
  threshold_z: 1.0
  spread_buffer_m: 5
  match_iou: 0.1
trajectory:
  min_persistence: 2
  recovery_fraction: 0.2
outputs: {cog: true, gpkg: true, report: html}
```

CLI

```
szcd ingest     --config config/season.yaml --date 2026-07-14
szcd preprocess --config config/season.yaml --date 2026-07-14
szcd index      --config config/season.yaml --date 2026-07-14
szcd stress     --config config/season.yaml --date 2026-07-14
szcd change     --config config/season.yaml --pair 2026-07-01:2026-07-14
szcd trajectory --config config/season.yaml --field F012
szcd report     --config config/season.yaml --run-id latest
szcd run        --config config/season.yaml            # all new flights end
to end
```

Common flags: `--field` to limit to fields, `--force` to ignore cache, `--workers N`, `--dry-run`. Exit code is non-zero if any flight fails a QA gate.

Raster conventions — float32 for indices and Δ, uint8 for masks and classes, nodata −9999 / 255, internal tiling 512 × 512, DEFLATE compression, overviews built. File names follow the directory layout above.

Python API

```
from szcd import Season
s = Season.from_yaml("config/season.yaml")
pair = s.change("F012", "2026-07-01", "2026-07-14")
pair.patches.query("event == 'expanding'").to_file("expanding.gpkg")
s.trajectories("F012").plot(metric="area_m2")
```

Validation, QA and limitations

Every run is gated on calibration, registration and noise checks; a flight failing any gate is excluded from change pairs rather than silently compared.

QA gates

Gate	Pass criterion	On fail
Panel calibration	Gain within ±15% of season median	Flight flagged, excluded
Co-registration	Check-point RMSE ⩽ 0.5 px	Retry with TPS; else exclude
Valid coverage	⩾ 85% of field pixels unmasked	Pair computed, flagged partial
PIF residual	Per-band RMSE ⩽ 0.02 reflectance	Warn; noise floor raised
Stable-area Δz	Median	Δz

Accuracy assessment — with scouting points, compute a confusion matrix of stressed vs healthy per date, and of change class per pair (overall accuracy, per-class precision/recall, kappa). Points within 1 m of a patch edge are reported separately because of registration uncertainty. Without scouting data, a stratified random sample of ⩾ 50 points per class should be photo-interpreted from the orthomosaics.

Known limitations

- Indices show *that* vigour changed, not *why*; cause attribution needs scouting or thermal/soil data.
- NDVI saturates above roughly LAI 3; late-season NDVI change underestimates stress.
- Senescence and harvest-adjacent dates produce field-wide declines; field normalization reduces but does not remove this.
- Wind-driven canopy motion and row-direction shadows cause edge noise at fine GSD.
- Sensor or firmware changes mid-season break comparability; re-baseline the reference date if unavoidable.

Roadmap

- [ ] Thermal band support (CWSI) for water-stress separation
- [ ] Satellite gap-fill (Sentinel-2) between flights at coarser resolution
- [ ] Probabilistic change maps with per-pixel confidence
- [ ] Web viewer for side-by-side date swiping and patch timelines